

USB Communication



File: usb.kicad_sch

Power Supply



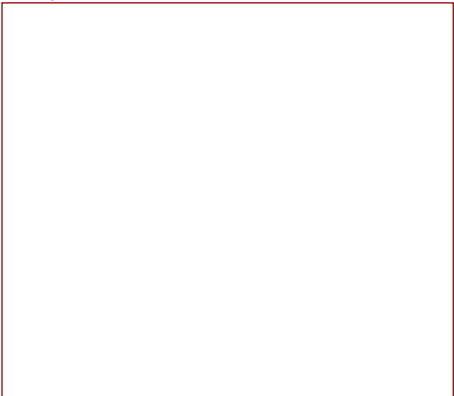
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MCU



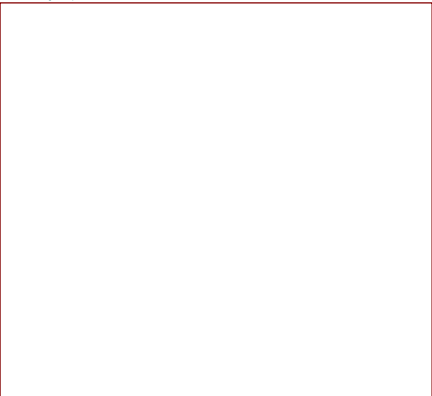
File: mcu.kicad_sch

Heating Element



File: heating_element.kicad_sch

Cooling System



File: cooling_system.kicad_sch

PCB Mounting Holes

H1
MountingHole

H2
MountingHole

H3
MountingHole

H4
MountingHole

PCB Fiducials

FID1
Fiducial

FID2
Fiducial

FID3
Fiducial

FID4
Fiducial

FID5
Fiducial

FID6
Fiducial

FID7
Fiducial

FID8
Fiducial

SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /
File: atech_reflow_plate_v2.kicad_sch

Title: ReflowDesk AT-MK1

Size: A3
KiCad E.D.A. 10.0.1

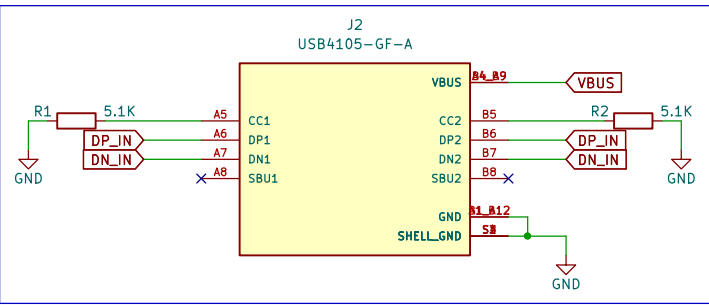
Date: 2026-04-30

Rev: v1.0

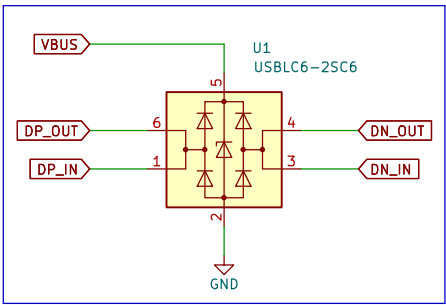
Id: 1/6



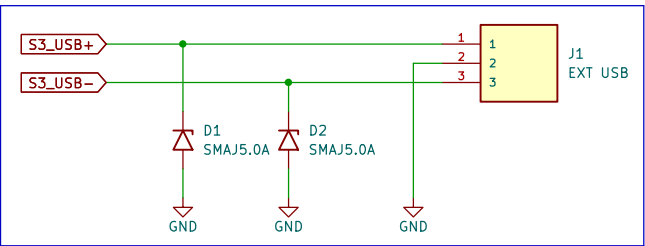
USB 3.1 Type-C Connector (Receptacle)



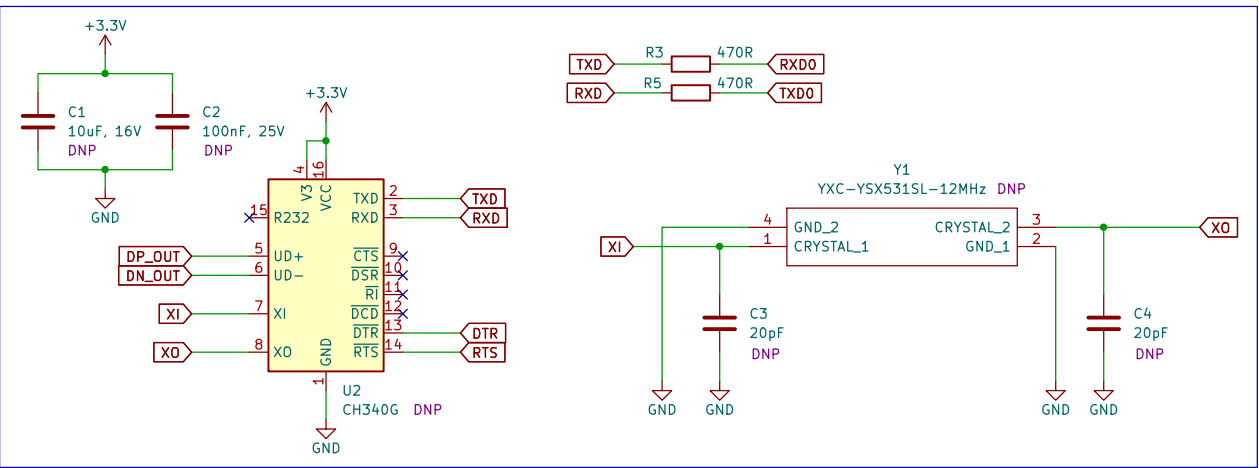
ESD Protection Circuit



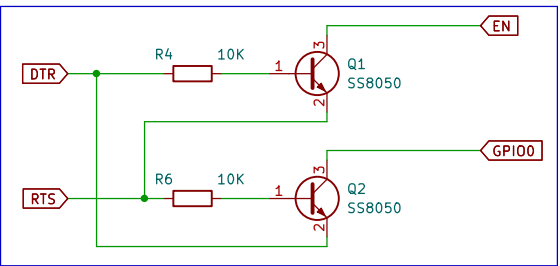
External USB Module Connector



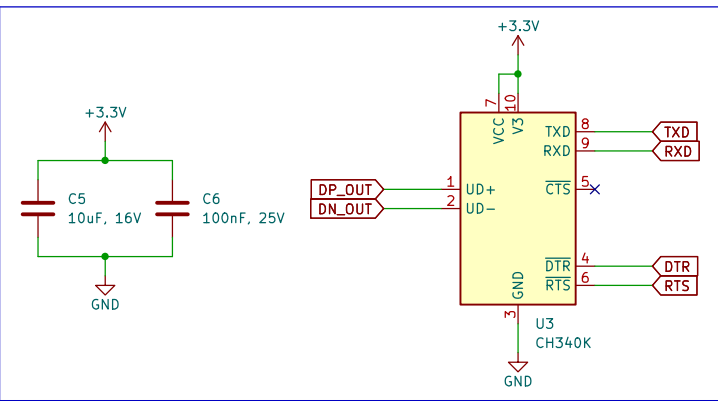
USB to UART Bridge Circuit (Option-2)



ESP32 MCU Auto Programming Circuit



USB to UART Bridge Circuit (Option-1)

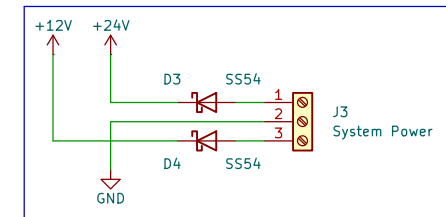


SMD Reflow Soldering Hot Plate
A-Tech Officials
Sheet: /USB Communication/
File: usb.kicad_sch

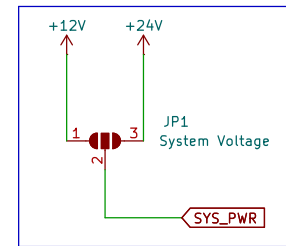
Title: ReflowDesk AT-MK1 (USB Communication)

Size: A3	Date: 2026-04-30	Rev: v1.0
KiCad E.D.A. 10.0.1		Id: 2/6

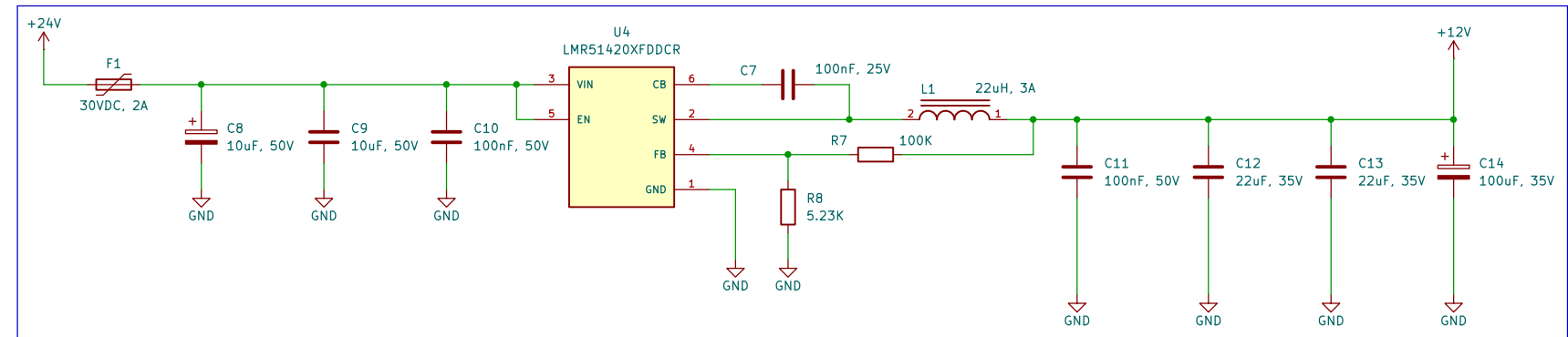
System Power Connector



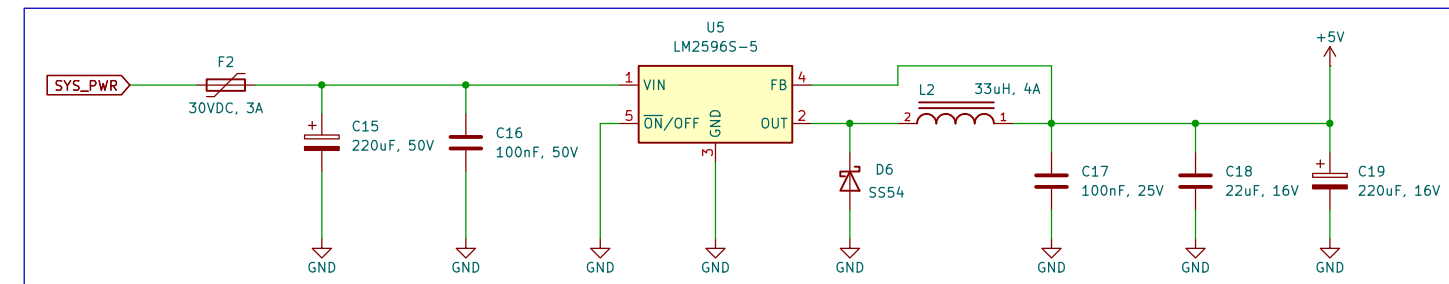
System Power Voltage Selection Jumper



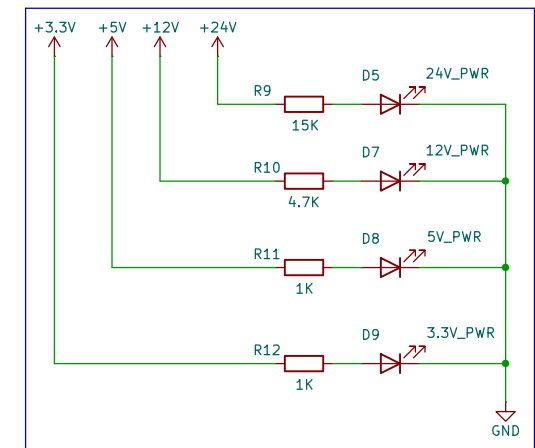
12V, 2A Buck Converter (PWM Fan Power)



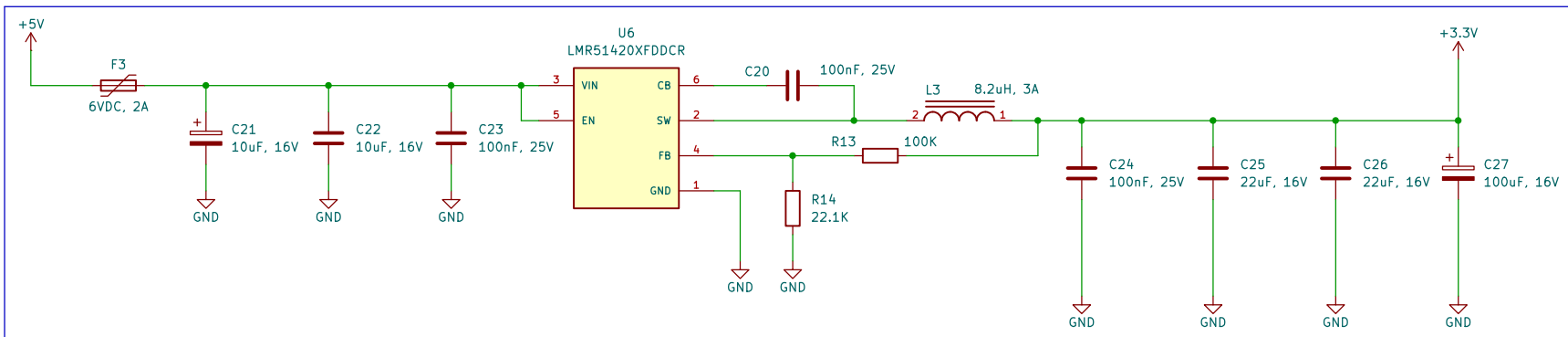
5V, 3A Step-Down Regulator/Buck Converter



System Power LED Indicators



3.3V, 2A Buck Converter (MCU Power + Peripherals Power)



SMD Reflow Soldering Hot Plate

A-Tech Officials

Sheet: /Power Supply/

File: power_supply.kicad_sch

Title: ReflowDesk AT-MK1 (Power Supply)

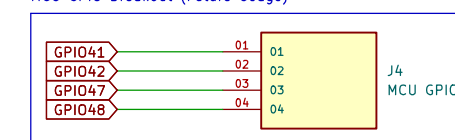
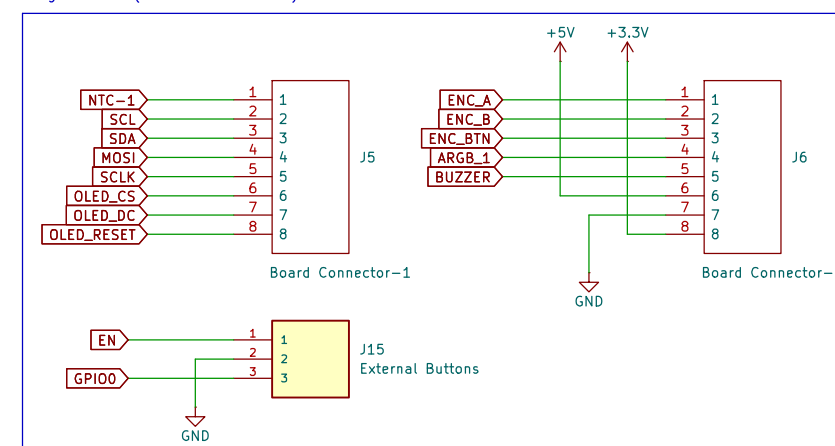
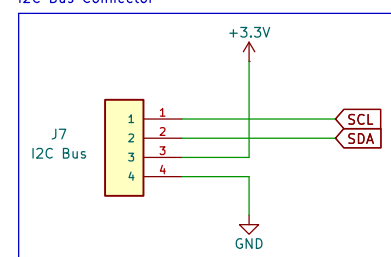
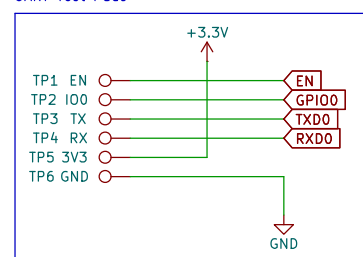
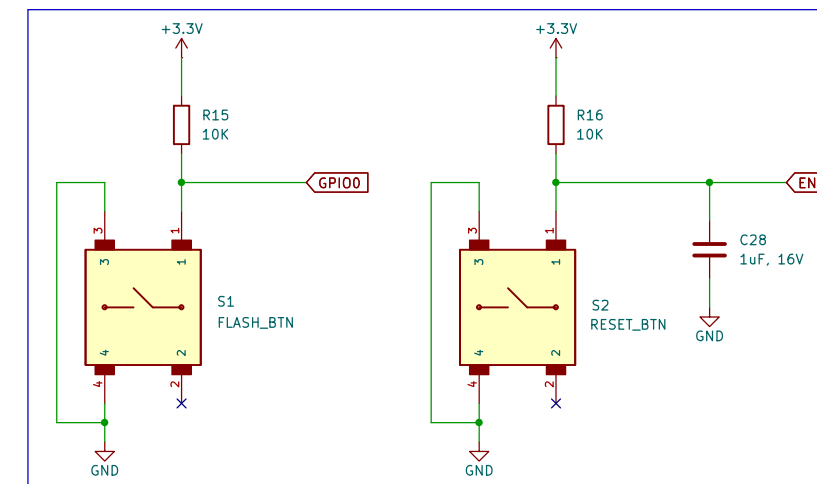
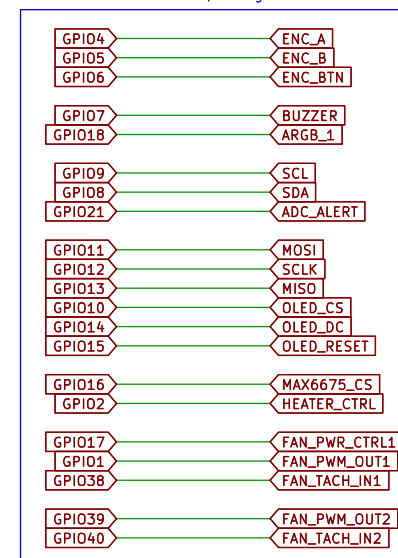
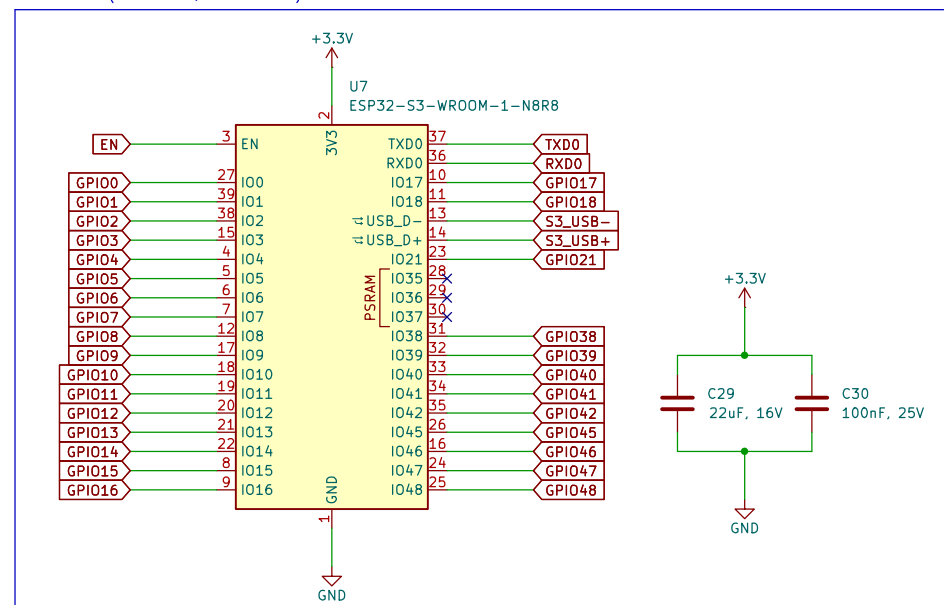
Size: A3

Date: 2026-04-30

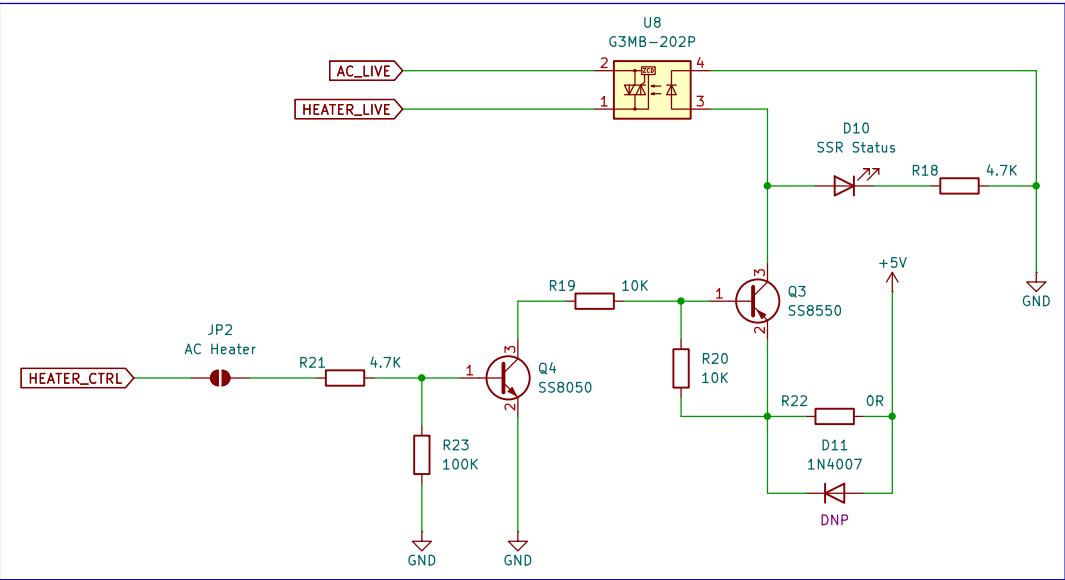
Rev: v1.0

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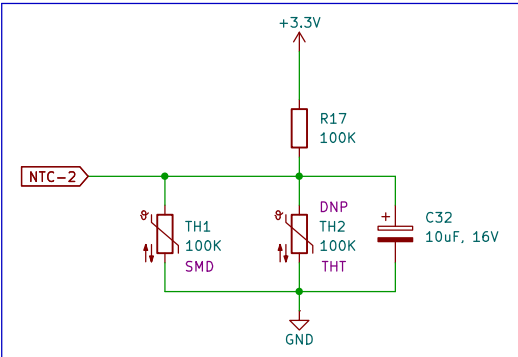
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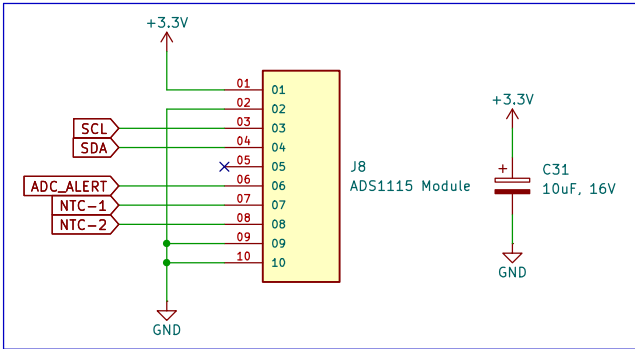
220VAC PTC Heating Element Control (Active-Low SSR, Active-High Signal)



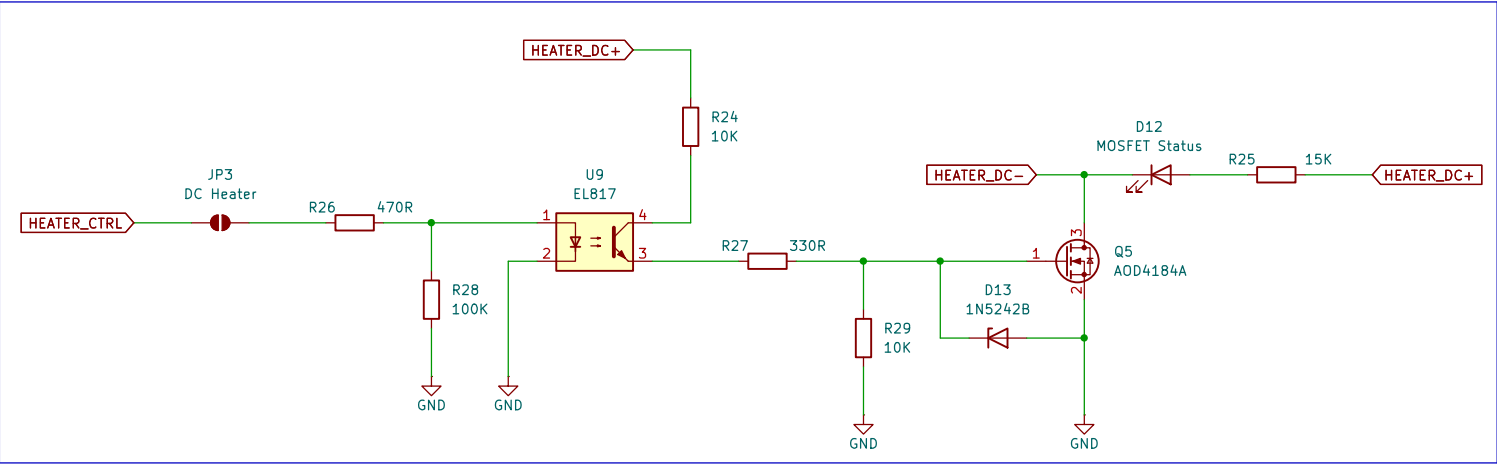
NTC Ambient Temp Sensor (Controller Board)



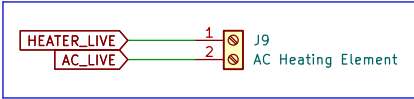
ADS1115 16-bit ADC Module Connector



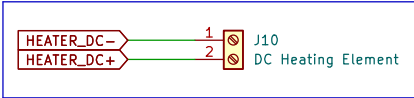
DC PTC Heating Element Control (Low-Side MOSFET Switch)



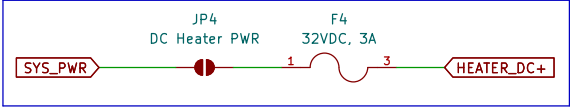
220VAC PTC Heating Element Connector



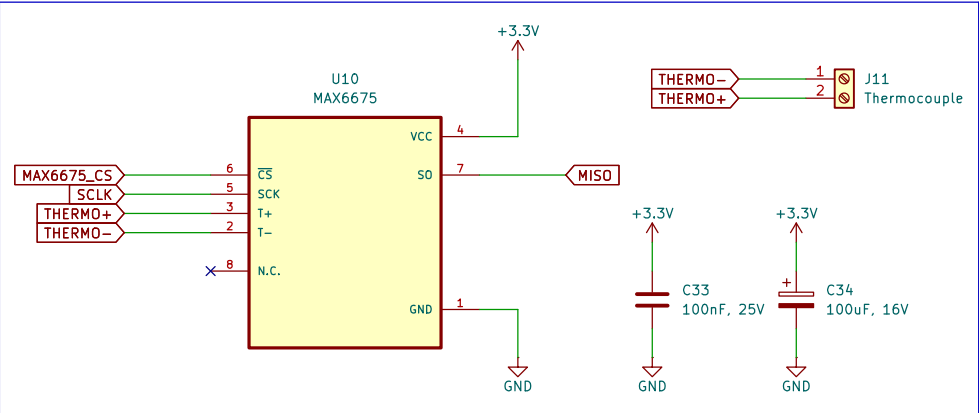
DC PTC Heating Element Connector



DC PTC Heating Element Power Enable Jumper



Hot Plate Temperature Sensor (K-Type Thermocouple)



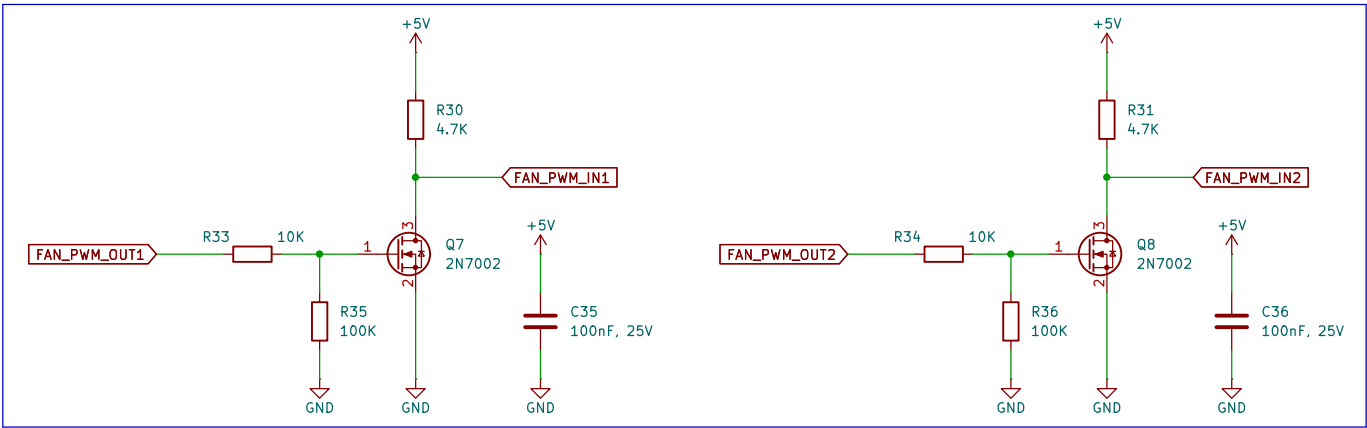
SMD Reflow Soldering Hot Plate
A-Tech Officials
Sheet: /Heating Element/
File: heating_element.kicad_sch

Title: ReflowDesk AT-MK1 (Heating Element)

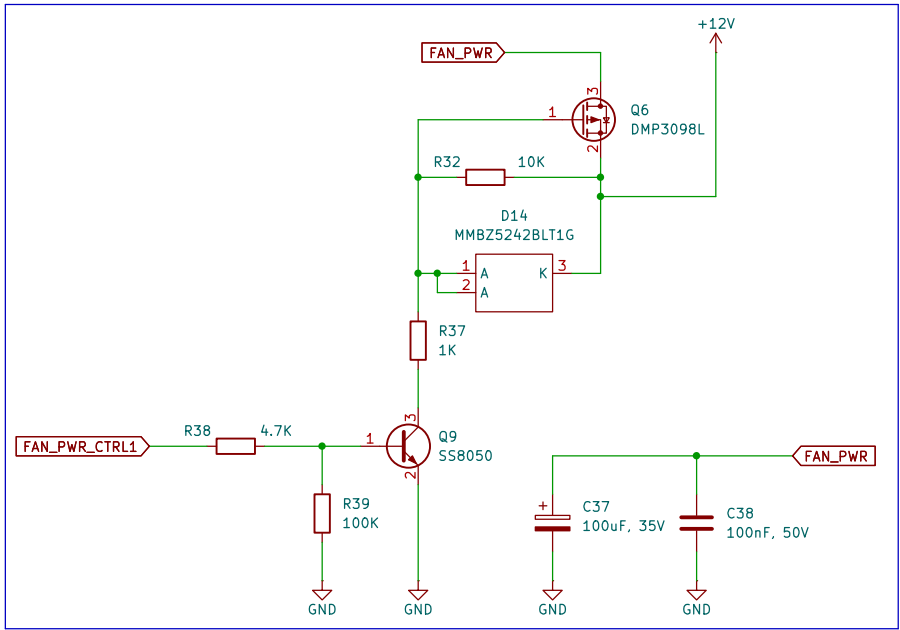
Size: A3 Date: 2026-04-30
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Rev: v1.0
Id: 6/6

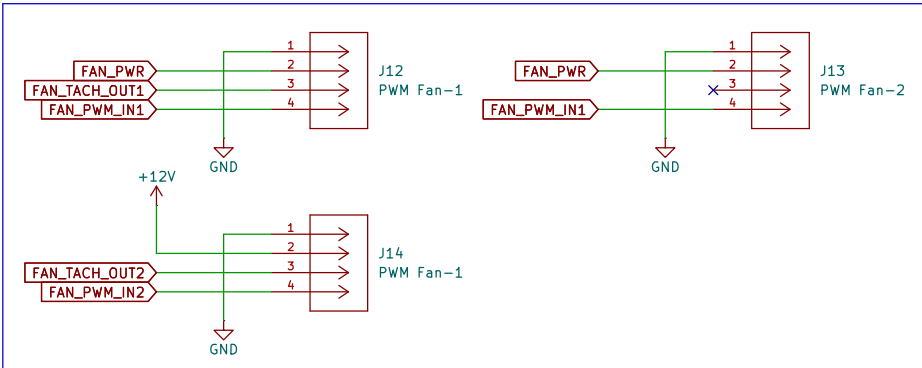
N-MOSFET Open-Drain Level Shifter / PWM Distribution Buffer



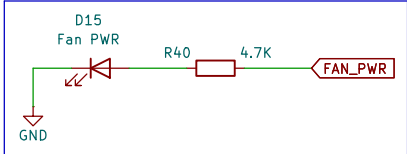
12VDC PWM Fan Power Control



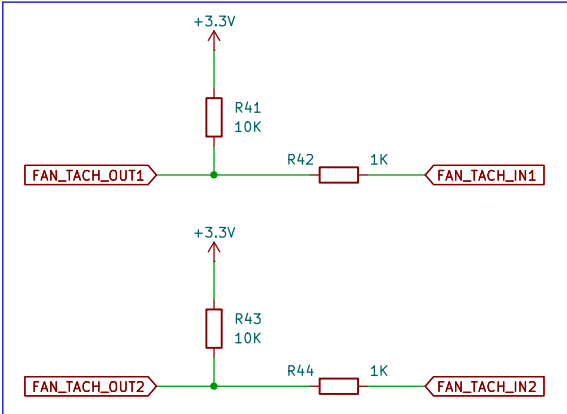
12VDC PWM Fan Connectors



Fan Power Status LED



Fan Tachometer Read Circuit



SMD Reflow Soldering Hot Plate
A-Tech Officials
Sheet: /Cooling System/
File: cooling_system.kicad_sch

Title: ReflowDesk AT-MK1 (Cooling System)

Size: A3 Date: 2026-04-30
KiCad E.D.A. 10.0.1

Rev: v1.0
Id: 7/6